

Lab 41 Worksheet — Network Attack Recognition — Injection, XSS and Data Leakage

Name: ____ Date: ____

Exam objective: Recognise common application and network attacks including SQL injection, cross-site scripting, on-path attacks and denial of service, and state the defence for each (Core 2 objective 2.4).

Record as you go

Step	What you did	What you observed
1	Open https://alfredang.github.io/cybersecuritysimulator/ and select the SQL Injection module. The login form and its data are entirely fake and in-memory.	
2	Enter a normal username and password and observe how the live SQL query display changes as you type.	
3	Enter the suggested demonstration input — username 'admin' with password ' OR '1'='1 — and record how the query structure changes and why it now always evaluates true.	
4	Record the SQL injection defence: parameterised queries and prepared statements, plus input validation, so that user input can never alter query structure.	
5	Switch to the XSS module, enter text into the input, and compare the unsafe rendering against the safe rendering. No code actually executes.	
6	Record the XSS mechanism: an attacker injects script into a page that other users then load, so the victim's browser executes it in the context of the trusted site.	

Step	What you did	What you observed
7	Record the XSS defence: output encoding, input validation and a Content Security Policy, and complete the module's quiz on prevention.	
8	Open the Data Leakage module, toggle the six security practices and record how the risk score responds to each.	
9	Identify from the risk score which single practice reduces risk most, and record the combination that produces the lowest score.	
10	Build the attack reference table covering SQL injection, XSS, denial of service, distributed denial of service, on-path or man-in-the-middle, DNS poisoning, ARP spoofing, zero-day and insider threat.	
11	For each attack record the mechanism, the observable symptom a technician would see in a ticket, and the defence.	
12	Record the A+ technician's scope explicitly: recognise the symptom, isolate the affected system, preserve evidence and escalate to security. Do not attempt to counter-attack or to investigate beyond your authorisation.	

Verification

Success criterion: Your table covers at least eight attacks with mechanism, symptom and defence; you correctly explain why ' OR '1'='1 bypasses authentication; the data leakage assessment names the highest-impact practice from the measured score; and the escalation boundary is stated.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
